Urban Mobility Intelligence System with Explainable AI and Event-Aware Analysis

DS504: Big Data Analytics

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Abstract

This project presents an Urban Mobility Intelligence System for analyzing New York City taxi movement patterns. The system integrates NYC TLC yellow taxi trip records, historical hourly weather data, and NYC public event datasets to analyze mobility behavior, predict taxi trip duration, and interpret model outputs using Explainable AI techniques. The implementation leverages Apache Spark for scalable data processing, Pandas for analytical summarization, XGBoost for predictive modeling, SHAP for model interpretability, and Folium for interactive geospatial visualization. The system processes approximately 2.9 million taxi trips and achieves strong predictive performance ($R^2 = 0.843$, MAE ≈ 176 seconds). Results indicate that trip distance, time of day, and spatial zones are the primary drivers of trip duration, while weather and event context provide additional insights. This work demonstrates the effectiveness of combining big data analytics with explainable AI to generate actionable insights for urban mobility planning.

Keywords

Urban mobility; NYC taxi; big data analytics; PySpark; XGBoost; SHAP; geospatial visualization; event-aware analysis; weather-aware prediction.

1. Introduction and Motivation

Urban transportation systems generate large volumes of operational data that can be leveraged to understand traffic patterns, travel-time variability, and mobility demand. In densely populated cities such as New York, taxi trip behavior is influenced by multiple factors including distance, location, time of day, weather conditions, and public events. For stakeholders such as commuters, transportation planners, and dispatch systems, it is essential not only to predict travel time but also to understand the underlying factors influencing these predictions.

The motivation of this project is to develop a practical and scalable mobility intelligence system that goes beyond static reporting. The system processes large-scale taxi data, integrates external contextual information, engineers meaningful features, builds predictive models, and generates interpretable outputs. Unlike traditional systems that focus solely on predictive accuracy, this project emphasizes explainability, enabling users to understand the key drivers of urban mobility patterns. This integration of big data processing, machine learning, explainable AI, and visualization provides a comprehensive end-to-end analytical solution.

Project goals

- Integrate heterogeneous datasets: taxi trips, weather, and public events.
- Create a scalable processing pipeline using Spark and Pandas.
- Engineer useful mobility features such as rush hour, airport trip flag, event level, and distance bucket.
- Train and evaluate an XGBoost model to predict taxi trip duration.
- Use SHAP explainability to identify which factors drive predictions.
- Generate map, chart, and prediction-demo outputs for final presentation.

Despite the availability of large-scale transportation data, most existing systems focus primarily on prediction of accuracy and lack interpretability. This limits their usefulness for real-world decision-making, where understanding the causes of congestion is as important as predicting it.

2. Dataset Description

The final project uses three aligned datasets. The taxi dataset serves as the primary mobility source, while weather and event datasets provide additional contextual information. A taxi-zone lookup file is also used to translate pickup and dropoff location IDs into readable zone names for analysis and reporting.

Dataset	Final data used	Role in project
NYC TLC Yellow Taxi Trip Records	February 2023 Parquet file with **2,913,955 rows**	Primary trip-level dataset containing timestamps, trip distance, fare details, pickup/dropoff zone IDs, and computed trip duration
Historical NYC Weather	**672 hourly observations** for February 2023	Provides hourly temperature, humidity, pressure, wind, and precipitation data aligned with trip pickup time
NYC Public Events	**13,576 event records** for February 2023	Adds citywide daily event counts and event-context features aligned with trip dates
TLC Taxi Zone Lookup	**265 zone records**	Maps numeric pickup/dropoff location IDs to readable borough and zone names for interpretation

A key limitation of the dataset is that the NYC TLC taxi data provides pickup and dropoff location IDs rather than exact latitude and longitude coordinates. As a result, precise street-level route reconstruction is not possible within this project. However, these location IDs still provide meaningful zone-level geographic information, which enables effective spatial analysis.

In future work, this limitation can be addressed by integrating taxi-zone polygon shapefiles, allowing more accurate geospatial visualization and analysis.

3. System Architecture

The system is designed as a **modular and scalable analytics pipeline**, where each module is responsible for a specific stage of data processing. The main execution script (`main.py`) orchestrates the complete workflow, ensuring smooth integration of all components. This modular design improves maintainability, simplifies debugging, and allows easy extension of the system with additional features or datasets.

File / module	Purpose
main.py	Orchestrates the complete pipeline including data loading, preprocessing, merging, feature engineering, modeling, explainability, and visualization
src/data_loading.py	Loads taxi, weather, and event data from the raw data folder.
src/preprocessing.py	Cleans data, standardizes column formats, parses timestamps, and prepares taxi, weather, and event datasets
src/merging.py	Merges taxi data with hourly weather data and daily event counts
src/feature_engineering.py	Creates model features such as hour, rush hour, distance bucket, airport trip flag, event level, and weather bands.
src/modeling.py	Trains and evaluate the XGBoost regression model.
src/shap_explainability.py	Generates SHAP-based feature importance plots and explanation visualizations
src/map_visualization.py	Creates the interactive Folium-based map showing traffic density and hotspots
src/analytics.py	Generates business insights, summary tables, and visualization charts
predict_trip.py	Provides a user-facing prediction interface for estimating trip duration based on input locations and time

Pipeline flow

The system follows a structured step-by-step pipeline:

1. **Data Loading**
   Load raw taxi, weather, event, and taxi-zone datasets
2. **Data Preprocessing**
   Clean timestamps, compute trip duration, remove invalid records, and normalize data formats
3. **Data Integration**
   Merge taxi data with weather (by pickup hour) and events (by pickup date)
4. **Feature Engineering**
   Create predictive features including temporal, spatial, and contextual variables
5. **Model Training**
   Train an XGBoost regression model to predict taxi trip duration

6. **Explainability**
   Apply SHAP to interpret feature importance and model behavior
7. **Visualization and Output**
   Generate interactive map visualizations, analytical charts, and prediction outputs

4. Key Technologies Used

The project leverages a combination of big data processing frameworks, machine learning models, and visualization tools to build a scalable and interpretable analytics pipeline.

Technology	Use in the project
Apache Spark / PySpark	Distributed data loading, preprocessing, and merging of large-scale taxi datasets
Pandas	Analytical processing, feature summarization, and generation of presentation-ready outputs
XGBoost	Gradient-boosted tree model used for accurate trip-duration prediction
SHAP	Explainable AI technique used to interpret feature importance and model behavior
Folium / Leaflet	Interactive map visualization for traffic density, congestion hotspots, and event overlays
Streamlit and CLI	User-facing interfaces for visualization and route-level prediction demonstration

5. Data Preprocessing and Feature Engineering

The preprocessing stage ensures that raw datasets are transformed into a clean and structured format suitable for analysis. This includes standardizing column names, converting pickup and dropoff timestamps into datetime format, computing trip duration, handling missing values, and removing invalid or noisy records. External datasets such as weather and events are also aligned temporally with taxi data.

The feature engineering stage transforms raw attributes into meaningful variables that improve model performance and interpretability. These features capture temporal patterns, trip characteristics, and contextual factors affecting urban mobility.Engineered features

Feature family	Examples	Why it matters
Temporal	hour, day of week, month, weekend flag, rush-hour flag	Captures regular daily and weekly mobility patterns.

Feature family	Examples	Why it matters
Trip structure	trip_distance, distance_bucket, pickup/dropoff TLC zones	Represents the physical and zone-based shape of the trip.
Context	event_count, event_level, temperature, wind, humidity, pressure, precipitation	Adds external conditions aligned to the pickup time/date.
Special route type	is_airport_trip	Separates JFK, LaGuardia, and Newark airport trips from normal city trips.

Explanation of two important engineered features

The **distance_bucket** feature groups continuous trip distance into interpretable categories such as very short, short/medium, long city trips, and very long trips. While the model uses exact distance values, this additional categorization helps capture different types of travel behavior and improves model interpretability.

The **is_airport_trip** feature is a binary indicator that identifies whether a trip involves major airports such as JFK, LaGuardia, or Newark. Airport trips typically have longer distances, different routing patterns, and distinct traffic characteristics compared to regular urban trips. Including this feature helps the model better differentiate between these patterns.

Outlier handling

To improve model accuracy and reliability, the preprocessing pipeline removes invalid or extreme records before training. This includes filtering trips with unrealistic durations, zero or incorrect distances, abnormal speeds, and missing or unknown zone identifiers. By eliminating noisy data, the model is trained on realistic mobility patterns, which significantly improves prediction performance and stability.

6. Analytical Insights

In addition to predictive modeling, the project generates **business-oriented analytical insights** stored in the `outputs/analytics/insights` directory. These insights transform technical results into **interpretable and actionable findings**, covering key aspects such as time-based mobility patterns, event impact, weather influence, travel corridors, and airport trip behavior.

6.1 Time-based mobility

The analysis reveals clear temporal patterns in urban mobility. The **slowest hour** occurs at **14:00 (2 PM)** with an average trip duration of **18.7 minutes**, while the **busiest hour** is **18:00 (6 PM)**, indicating peak demand during evening hours.

This demonstrates that **time of day is a strong operational factor** influencing both traffic congestion and travel demand. These patterns can be used by travelers, ride-hailing services, and transportation planners to optimize travel schedules and reduce delays.

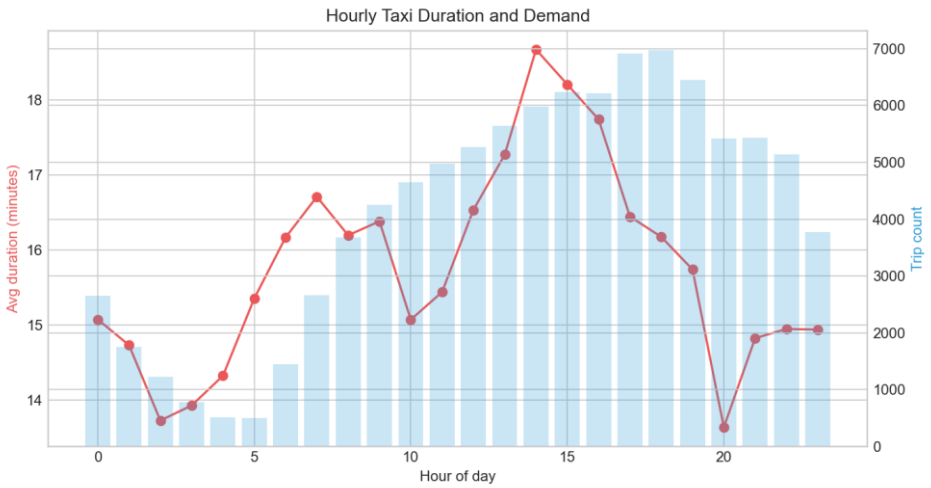


Figure 1. Hourly mobility analysis shows average duration and trip volume by hour.

6.2 Event impact

The analysis indicates that **citywide event counts have a limited direct impact on overall trip duration**. This suggests that using aggregated event data may not sufficiently capture localized traffic disruptions.

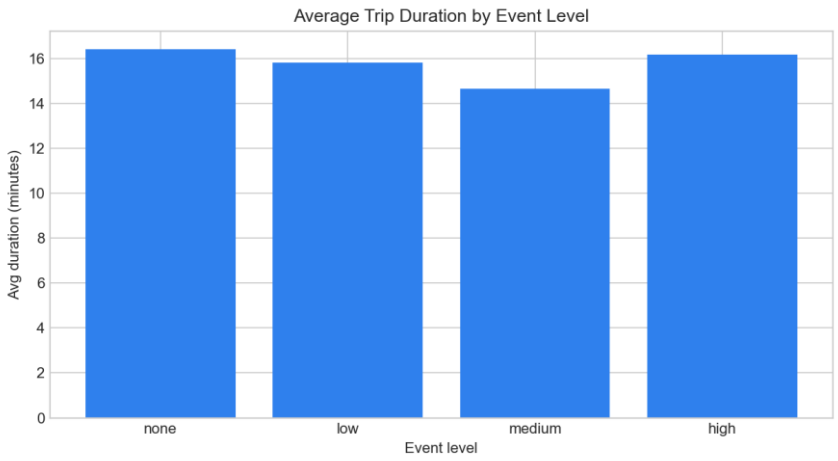


Figure 2. Average trip duration by event level.

6.3 Weather impact

Weather conditions influence trip duration; however, their effect is **relatively smaller compared to core mobility factors** such as distance, time of day, and location. The analysis shows that the **slowest temperature band (mild conditions)** results in an average duration of **17.6 minutes**, with only moderate variation across other temperature ranges.

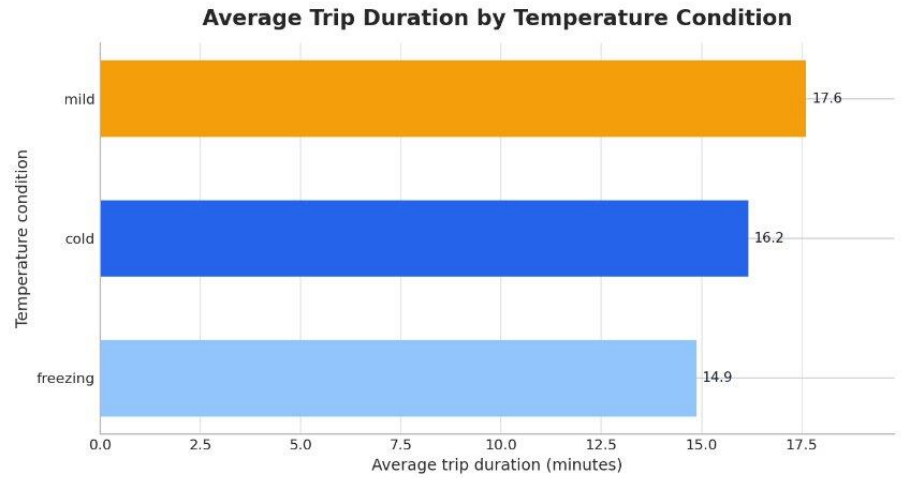


Figure 3. Average trip duration by temperature band.

6.4 Common travel corridors

The analysis identifies high-frequency travel routes, with the **most common corridor** being from **Upper East Side South to Upper East Side North**, accounting for **701 trips** in the dataset.

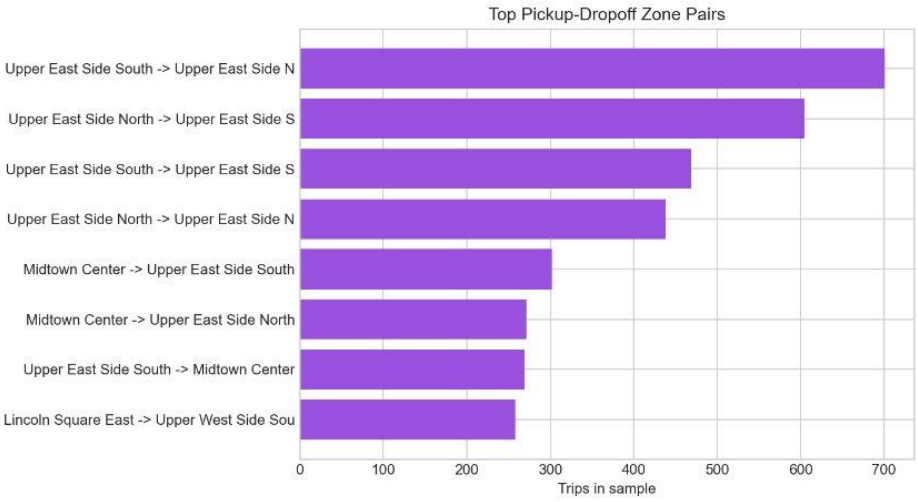


Figure 4. Most frequent pickup-dropoff zone pairs after filtering unknown zones.

6.5 Airport trip behavior

Airport trips exhibit significantly different characteristics compared to regular city trips. On average, **airport trips last 35.5 minutes and cover 14.1 miles**, whereas **non-airport trips average 14.0 minutes and 2.3 miles**.

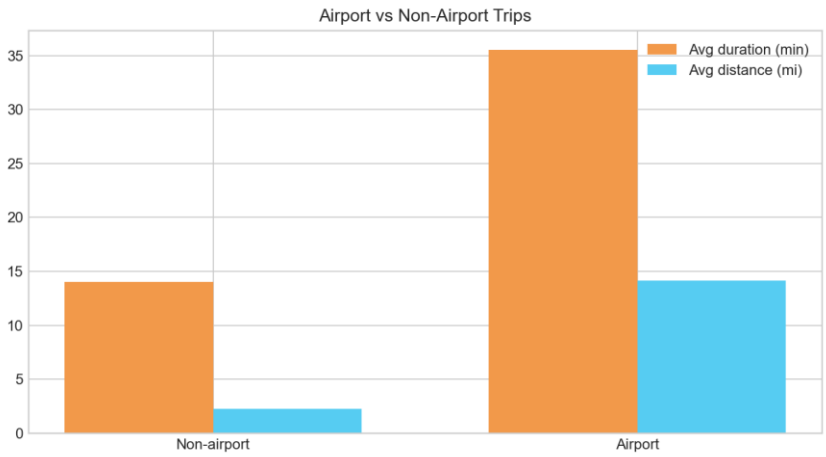


Figure 5. Airport trips are longer and behave differently from normal city trips.

7. Predictive Modeling and Evaluation

The project uses an **XGBoost regression model** to predict taxi trip duration (`trip_duration_sec`). XGBoost was selected due to its strong performance on structured tabular data, ability to capture nonlinear relationships, robustness to feature scaling, and seamless integration with SHAP-based explainability.

The model is designed using **pre-trip features only**, excluding post-trip attributes such as `total_amount` or `tip_amount`. This ensures that the system remains realistic and applicable to real-world scenarios, where predictions must be made **before a trip is completed**.

Metric	Final value	Interpretation
Taxi trips loaded	2,913,955 rows	NYC TLC yellow taxi trip records in Parquet format
Modeling sample	48,305 valid rows	Data after preprocessing, feature engineering, and outlier removal
Weather records	672 hourly rows	Hourly weather variables for February 2023
Event records	13,576 events	NYC Open Data event records aligned by trip date
Model result	**R2 = 0.843** **MAE = 176 sec**	**Strong predictive performance for trip duration estimation**

The final model performance is substantially stronger than the early baseline. After feature refinement and outlier filtering, the model achieved R2 = 0.843 and MAE = 176 seconds. This means the model explains approximately 84 percent of the test-set variation in trip duration and is typically within about 2.9 minutes of the actual trip duration.

Model features

The final model demonstrates strong predictive capability, achieving an **R^2 score of 0.843** and a **mean absolute error (MAE) of approximately 176 seconds**. This indicates that the model explains

approximately **84% of the variance** in trip duration and typically predicts travel time within **about 2.9 minutes** of the actual value.

8. Explainable AI with SHAP

SHAP (SHapley Additive exPlanations) is used to interpret the XGBoost model and provide insight into feature contributions. SHAP assigns each feature a contribution value for every prediction, enabling both **global interpretation (feature importance)** and **local interpretation (directional impact on individual predictions)**. This makes the model more transparent and easier to understand.

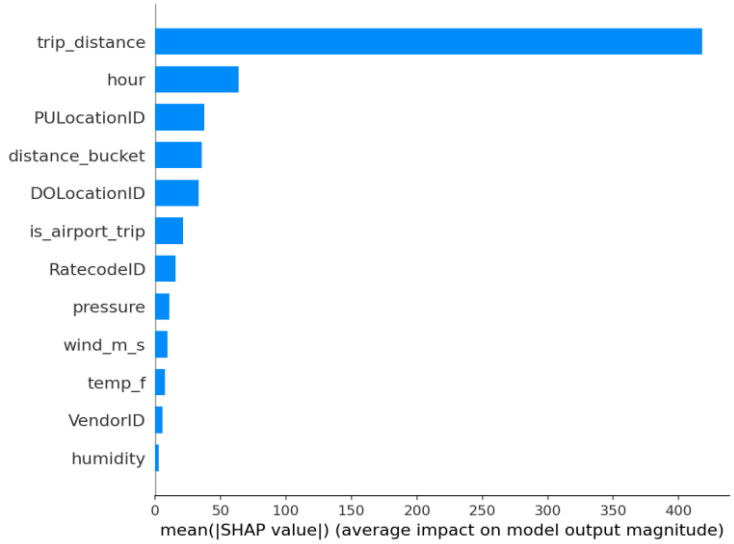


Figure 6. SHAP bar plot ranking the average impact of each feature.

The SHAP summary bar plot indicates that **trip_distance is the most influential feature**, which aligns with real-world expectations, as longer trips generally require more time. Other important features include **hour of day, pickup location ID, distance bucket, dropoff location ID, and the airport-trip indicator**.

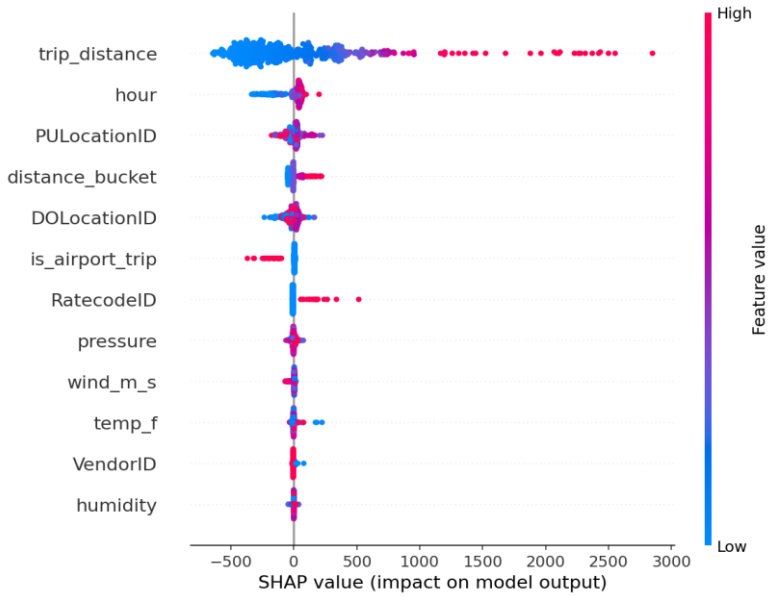


Figure 7. SHAP beeswarm plot showing how high and low feature values push predictions up or down.

The SHAP beeswarm plot provides a more detailed view of feature behavior. High values of **trip_distance** push predictions toward higher travel times, while lower values reduce predicted duration. Similarly, the **hour feature** demonstrates that time of day significantly impacts travel time due to varying traffic conditions.

The **airport-trip indicator** also contributes strongly to predictions, reflecting the distinct characteristics of airport routes, which typically involve longer distances and different traffic patterns. This observation is consistent with the analytical insights, where airport trips were found to have significantly higher average duration and distance compared to regular city trips.

Overall, the SHAP analysis validates the model by aligning with real-world mobility behavior, thereby **increasing trust and interpretability of the predictive system**.

9. Interactive Map and Demonstration

The project includes an interactive map visualization that provides an intuitive and user-friendly interface for exploring urban mobility patterns. The map, generated as `urban_mobility_map.html`, displays **traffic density, congestion hotspots, and event markers**, along with a congestion-severity legend for better interpretation.

The visualization can be directly opened in a web browser and is designed for use during demonstrations. Due to the use of pickup and dropoff zone IDs instead of precise geographic coordinates, the map represents **zone-level mobility patterns** rather than exact street-level routes. Despite this limitation, it effectively conveys spatial trends and traffic intensity across the city.

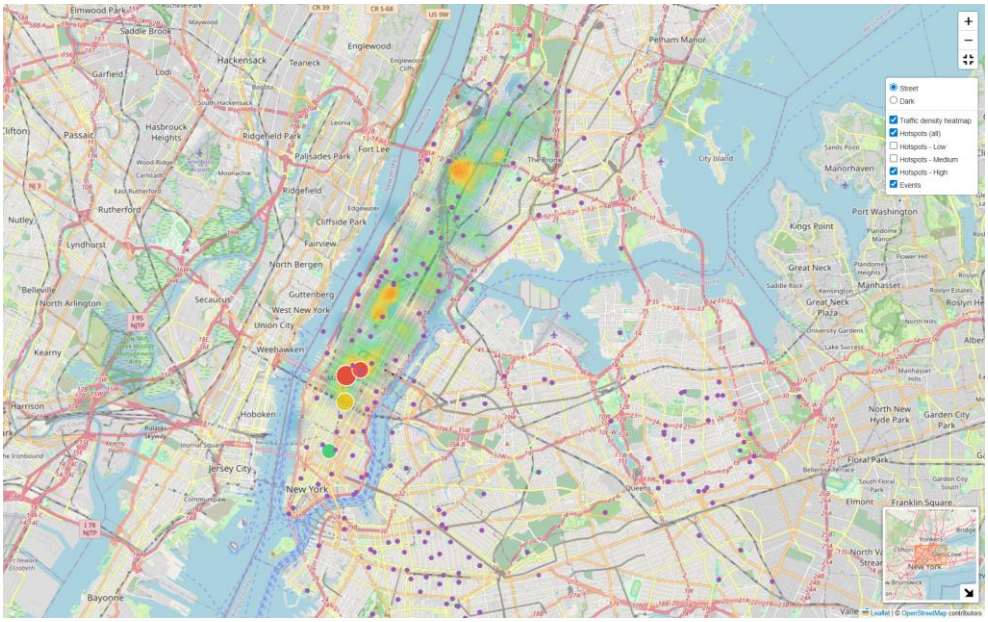


Figure 8. Interactive map output showing NYC mobility density, hotspots, event markers, and layer controls

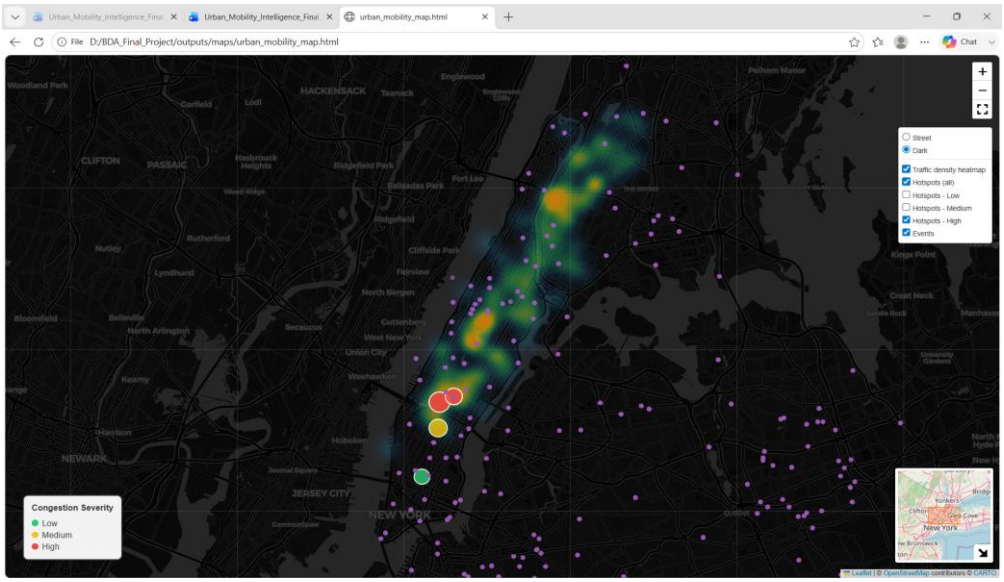


Figure 9. Interactive map output showing NYC mobility density, hotspots, event markers, and layer controls.

Prediction Demonstration

In addition to visualization, the system includes a **user-facing command-line prediction interface** that allows users to estimate trip duration based on input locations and time.

For example, the following command predicts a trip from Times Square to JFK Airport at 5:30 PM on February 10, 2023:

```
(.venv) (base) PS D:\BDA_Final_Project> python predict_trip.py --pickup "Times Square" --dropoff "JFK Airport" --datetime "2023-02-10 17:30"

Urban Mobility Prediction
-------------------------
Pickup:  Times Square (40.758, -73.9855)
Dropoff: JFK Airport (40.6413, -73.7781)
Date/time: 2023-02-10 17:30
Approx distance used: 18.26 miles
Events on that date: 511
Estimated duration: 49.7 minutes
Estimated congestion: Low
```

The system predicts a trip distance of **18.26 miles**, with **511 events** occurring on that date, resulting in an estimated travel time of **49.7 minutes** with low congestion. Additional test cases, such as trips from Grand Central to Wall Street and Penn Station to LaGuardia Airport, produced varied predictions, demonstrating that the system dynamically adapts to different routes and temporal conditions. This interactive prediction capability highlights the practical applicability of the model and reinforces its usefulness as a decision-support tool.

```
(.venv) (base) PS D:\BDA_Final_Project> python predict_trip.py --pickup "Grand Central" --dropoff "Wall Street" --datetime "2023-02-10 09:00"

Urban Mobility Prediction
-------------------------
Pickup:  Grand Central (40.7527, -73.9772)
Dropoff: Wall Street (40.706, -74.0086)
Date/time: 2023-02-10 09:00
Approx distance used: 4.89 miles
Events on that date: 511
Estimated duration: 24.8 minutes
Estimated congestion: Medium
```

10. Effectiveness of the Solution

The proposed solution is effective as it successfully completes the entire pipeline from **raw data processing to actionable insights**. The system processes large-scale taxi data, integrates external weather and event context, performs feature engineering, builds a predictive model, and generates interpretable outputs through explainable AI and visualization.

The model performance improved significantly after refined feature engineering and robust outlier handling, demonstrating that the system is not only technically functional but also **analytically meaningful and reliable**. By combining prediction, explanation, and visualization, the solution provides a comprehensive framework for understanding urban mobility patterns.

Strengths

- End-to-end pipeline that can be executed reproducibly through `main.py`
- Integration of real-world datasets aligned by time and context
- Strong predictive performance (**$R^2 = 0.843$**, **MAE ≈ 175.7 seconds**)
- SHAP-based explainability providing transparent model interpretation
- Business-oriented analytical insights that are understandable to non-technical stakeholders
- Interactive map visualization for intuitive exploration of traffic patterns
- Prediction interface enabling practical, real-time use of the model

Limitations

- The TLC dataset does not provide exact pickup and dropoff latitude/longitude coordinates, limiting street-level route analysis
- Event impact is currently modeled using **citywide daily event counts**, which may not capture localized congestion effects
- Zone IDs are treated as numerical features; more advanced spatial encoding methods could improve modeling
- The map visualization is limited to zone-level representation and does not include detailed geospatial polygons

These limitations primarily arise from **data granularity constraints rather than model design**, and therefore represent opportunities for further enhancement.

11. Future Work

Future improvements can significantly enhance both the analytical depth and practical applicability of the system. One key enhancement is the integration of **taxi-zone polygon shapefiles**, which would enable precise geospatial visualization and more accurate spatial analysis.

Another important extension is the development of **localized event modeling**, where event influence is measured based on proximity to pickup or dropoff locations rather than citywide aggregation.

Additional future work includes:

- Incorporating **real-time data streams** using streaming frameworks
- Applying **advanced machine learning models** for improved prediction accuracy

- Enhancing spatial feature representation using **geospatial embeddings or clustering techniques**

These improvements would transform the system into a **fully deployable urban mobility intelligence platform**.

12. Conclusion

This project demonstrates the effectiveness of integrating **big data analytics, machine learning, and explainable AI** to build a scalable and interpretable urban mobility system. The solution achieves strong predictive performance while also providing meaningful insights into the factors influencing traffic patterns.

By combining multi-source data integration, predictive modeling, explainability, and interactive visualization, the project highlights the potential of data-driven approaches to support **smarter transportation planning and decision-making**.

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